TransportIDF — Réseau de Transport d'Île-de-France

Part 2: Ticketing, Ridership and Quality of Service

Internal Document — Commercial & Customer Experience Department

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Table of Contents

1. Ticketing System Overview
2. Fare Products and Pricing
3. Validation and Revenue Tracking
4. Ridership and Traffic Analytics
5. Passenger Surveys and Satisfaction
6. Accessibility and Universal Design

1. Ticketing System Overview

The Île-de-France transport network uses a unified ticketing ecosystem based on the **Navigo** contactless smartcard system, managed by Île-de-France Mobilités. The system supports multiple fare products and is interoperable across all modes (metro, RER, bus, tramway, and suburban rail).

Ticketing Architecture

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                  ┌──────────────────────┐
                  │  Île-de-France       │
                  │  Mobilités (IDFM)    │
                  │  Fare Authority      │
                  └──────────┬───────────┘
                             │
          ┌──────────────────┼──────────────────┐
          ▼                  ▼                  ▼
   ┌──────────┐       ┌──────────┐       ┌──────────┐
   │  Navigo  │       │  Navigo  │       │ Ticket   │
   │  Pass    │       │  Easy    │       │ t+       │
   │ (monthly │       │(recharge │       │ (single  │
   │  annual) │       │  card)   │       │  paper)  │
   └──────────┘       └──────────┘       └──────────┘
         │                  │                  │
         └──────────────────┼──────────────────┘
                            ▼
              ┌──────────────────────┐
              │  Gate / Validator    │
              │  (station entry)     │
              └──────────────────────┘
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Key components:

- **Navigo Pass** — Contactless smartcard for subscription products (monthly, annual, weekly)
- **Navigo Easy** — Rechargeable card for pay-per-ride products
- **Ticket t+** — Single-ride magnetic ticket (being phased out in favor of contactless)
- **Smartphones** — NFC-enabled phones for ticket storage via the Île-de-France Mobilités app

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2. Fare Products and Pricing

The dataset models **15 fare products** covering the full range of travel needs:

Subscription Products

Product	Type	Price (EUR)	Valid Zones	Target
Navigo Mois	Monthly pass	86.40	All zones	Regular commuters
Navigo Annuel	Annual pass	950.40	All zones	Daily commuters (10% savings)

Navigo Semaine	Weekly pass	30.75	All zones	Short-term commuters
Navigo Jour	Day pass	8.65	All zones	Occasional full-day travel
Imagine R	Student annual	380.00	All zones	Students under 26

Pay-per-Ride Products

Product	Type	Price (EUR)	Valid Zones	Target
Ticket t+	Single ride	2.15	1–2	Individual metro/bus trip
Carnet 10 t+	10-ride pack	16.90	1–2	Regular single-ride users
Navigo Easy	Rechargeable card	2.00 (card only)	Variable	All pay-per-ride products

Tourist and Special Products

Product	Type	Price (EUR)	Valid Zones	Target
Paris Visite 1j	Tourist 1-day	13.95	1–3	Visitors
Paris Visite 2j	Tourist 2-day	22.65	1–3	Visitors
Paris Visite 5j	Tourist 5-day	43.30	1–5	Visitors (incl. airports)
Ticket Aéroport	Airport single	11.45	1–5	Airport travellers

Social Fare Products

Product	Type	Price (EUR)	Target
Ticket Jeunes WE	Youth weekend	4.60	Under-26 on Sat/Sun
Améthyste	Senior reduced	24.00/ month	Seniors 65+
Solidarité Transport	Social reduced	21.25/ month	Low-income residents

Pricing Policy

Île-de-France Mobilités adopted a **flat-fare zone policy** in 2015, where the Navigo monthly and annual passes cover **all zones** at a single price. This was a major shift from the previous zone-based pricing and resulted in significantly increased ridership in outer zones.

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3. Validation and Revenue Tracking

Every journey begins with a **ticket validation** — a tap-in event at a station gate or bus validator.

Validation Record Structure

Field	Description
Validation ID	Unique event identifier
Station ID	Where the validation occurred
Ticket Type ID	Which fare product was used
Validation Datetime	Exact timestamp of the tap
Gate ID	Physical gate or validator device
Card ID	Anonymised Navigo card number
Is Entry	True for entry, False for exit (on applicable systems)

Validation Volume

Metric	Value
Dataset validations	2,000 sample records
Real-world daily validations	~8 million
Annual validations	~2.5 billion
Peak hour share	35–40% of daily total

Revenue Analysis Dimensions

Validation data enables multi-dimensional revenue analysis:

- **By station** — Identify highest-revenue stations for commercial partnerships

- **By ticket type** — Monitor subscription vs. pay-per-ride mix
- **By time period** — Peak vs. off-peak revenue distribution
- **By zone** — Geographic revenue patterns
- **By card** — Frequency analysis for loyalty and pricing optimization

Fraud and Revenue Protection

The validation system also supports revenue protection:

- **Gate passage without validation** — Detected via passenger counts vs. validation counts
- **Expired passes** — Rejected at the gate with code logged
- [redacted] — Detected when exit zone exceeds ticket validity

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4. Ridership and Traffic Analytics

Daily Traffic Monitoring

The **traffic_daily** table records daily ridership at the line level:

Field	Description
Line ID	Which line was measured
Traffic Date	Date of measurement
Total Ridership	Total passengers for the day
Peak Hour Ridership	Passengers during morning and evening rush
Off-Peak Ridership	Passengers outside rush hours
[redacted]	
Day Type	Weekday, Saturday, or Sunday

Ridership Patterns

Pattern	Weekday	Saturday	Sunday
Total ridership index	100%	55–65%	35–45%
Morning peak (7:00–9:30)	30% of daily	10%	8%
Evening peak (17:00–19:30)	25% of daily	12%	10%
Off-peak daytime	30% of daily	55%	55%

Evening/night	15% of daily	23%	27%

Seasonal Variations

Period	Index vs. Average
January–February	95%
March–June	105%
July–August	70% (summer holidays)
September	110% (rentrée)
October–November	100%
December	85% (holidays)

Key Ridership Metrics

KPI	Target	Description
Annual ridership growth	+2%	Year-on-year passenger increase
Peak load factor	< 4 pax/m²	Comfort standard during rush hour
Off-peak utilisation	> 40%	Revenue optimisation target
Weekend ridership share	> 25%	Leisure travel growth indicator

5. Passenger Surveys and Satisfaction

Survey Programme

TransportIDF conducts systematic passenger satisfaction surveys across the network:

Field	Description
Line and Station	Where the survey was conducted
Survey Date	Date of data collection
Aspect	Dimension being evaluated

Rating	Score from 1 (very poor) to 10 (excellent)
Comment	Free-text respondent feedback
Age Group	Respondent demographic (18–24, 25–34, …, 65+)
Commute Frequency	Daily, weekly, occasional, first-time

Satisfaction Dimensions

Aspect	Description	Avg. Rating
Punctuality	On-time performance and reliability	6.5/10
Cleanliness	Vehicles and station cleanliness	5.8/10
Safety	Personal security feeling	6.2/10
Information	Real-time displays, announcements	6.0/10
Comfort	Seating, temperature, noise	5.5/10
Accessibility	Ease of use for reduced-mobility passengers	5.2/10
Frequency	Service frequency satisfaction	6.8/10
Crowding	Comfort during peak hours	4.5/10

Demographic Insights

Age Group	Avg. Satisfaction	Top Concern
18–24	6.1	Frequency, Crowding
25–34	5.8	Punctuality, Crowding
35–44	5.6	Accessibility, Information
45–54	5.9	Comfort, Safety
55–64	6.0	Accessibility, Cleanliness
65+	6.3	Accessibility, Safety

With **400 survey records**, this dataset supports satisfaction trend analysis, priority identification, and investment planning for service improvement.

6. Accessibility and Universal Design

Legal Framework

French law (*Loi du 11 février 2005*) mandates full accessibility of public transport for persons with reduced mobility (PRM). TransportIDF tracks accessibility equipment at every station:

Equipment Types

Equipment	Description	Typical Location
Elevator	Vertical transport between levels	Main entrance, platform
Escalator	Moving stairway	Mezzanine, exits
Tactile Strip	Ground-level guidance for visually impaired	Platform edge
Audio Announcement	Voice announcements at platforms and in vehicles	Station-wide
Wheelchair Ramp	Deployable ramp for level boarding	Platform level
Induction Loop	Hearing aid compatible audio	Ticket office, information desk

Equipment Record Structure

Field	Description
Equipment ID	Unique identifier
Station ID	Where the equipment is installed
Equipment Type	Category (elevator, escalator, etc.)
Location Description	Specific location within the station
Installation Date	When installed
Last Inspection Date	Most recent maintenance check
Status	Operational, out of service, or under repair
Manufacturer	Otis, Schindler, Kone, ThyssenKrupp, Mitsubishi

Accessibility KPIs

KPI	Target	Current
Stations with elevator access	100%	~65% (metro), 100% (tramway)
Equipment availability rate	> 97%	94.5%
Mean time to repair	< 48 hours	52 hours
Audio announcement coverage	100%	98%

With **180 equipment records**, this dataset enables accessibility compliance monitoring, maintenance prioritisation, and investment planning for universal access.

This document is Part 2 of the TransportIDF business description series. See Part 1 (Network Infrastructure & Operations) for network details and Part 3 (Maintenance, Incidents & Data Integration) for operational resilience coverage.